Training Objective Derivation for RetroDiff

The RetroDiff training objective follows the standard evidence lower bound (ELBO) framework for latent-variable diffusion models, then extends it to the retrieval-augmented setting. We derive the three-step chain below.

Step 1: Variational lower bound. Let $q(x_{1:T} \mid x)$ be the fixed forward (noising) process that gradually corrupts a clean sample x over T steps, and let $p_\theta(x_{0:T})$ be the learned joint generative model. Because the log-likelihood $\log p_\theta(x)$ is intractable, we bound it from below via Jensen's inequality applied to the concave log:

$$\log p_\theta(x) \;\geq\; \mathbb{E}_{q(x_{1:T}|x)}\left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)}\right] \;=\; -\mathcal{L}_{\text{ELBO}} \tag{1}$$

The inequality is tight when q matches the true posterior $p_\theta(x_{1:T} \mid x_0)$; in practice optimising $-\mathcal{L}_{\text{ELBO}}$ drives the model towards high likelihood without evaluating an intractable normaliser.

Step 2: Reconstruction–KL decomposition. Expanding the ELBO by Markov factorisation of both p_θ and q along the diffusion chain yields a reconstruction term at $t=1$ plus one KL divergence per denoising step:

$$\mathcal{L}_{\text{ELBO}} \;=\; \underbrace{\mathbb{E}_q\big[-\log p_\theta(x_0 \mid x_1)\big]}_{\text{reconstruction}} \;+\; \sum_{t=2}^{T} \underbrace{\mathrm{KL}\big(q(x_{t-1} \mid x_t, x) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)}_{\text{step-}t\text{ denoising}} \tag{2}$$

Each KL term measures how well the learned reverse step $p_\theta(x_{t-1} \mid x_t)$ matches the tractable forward posterior $q(x_{t-1} \mid x_t, x)$. Because the forward process has Gaussian transitions, both quantities are Gaussian and each KL reduces to a weighted ℓ_2 loss on the predicted noise, recoverable in closed form.

Step 3: Retrieval-augmented objective (RetroDiff contribution). Standard diffusion trains the reverse process without side information. RetroDiff augments each sample x with a set $R(x)$ of k retrieved exemplars drawn from a nearest-neighbour index over the training corpus. The per-sample ELBO is conditioned on these exemplars, and the outer expectation averages over the data distribution p_{data}:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) \;=\; \mathbb{E}_{x\sim p_{\text{data}}}\big[\mathcal{L}_{\text{ELBO}}(x \mid R(x))\big] \tag{3}$$

Conditioning the ELBO on $R(x)$ propagates retrieved appearance into every denoising KL term in (2): each reverse step $p_\theta(x_{t-1} \mid x_t, R(x))$ now has access to exemplar context. The reconstruction term in (1) similarly becomes $-\log p_\theta(x_0 \mid x_1, R(x))$. Minimising (3) therefore encourages the model to exploit retrieved structure across the entire denoising chain, rather than at a single conditioning step.